

DEPARTMENT OF MECHANICAL ENGINEERING

CURRICULUM OF M. Tech. (Mechanical - PLM)
(With effective from 2023-2024)

First Year					
Course Code	Name of the course	L	T	P	Credits
Semester I					
PCC-PL501	PLM Fundamentals	3	--	4	5
PCC-PL503	Data Management	3	--	2	4
PEC-PL504 to	Elective-I	3	--	2	4
PEC-PL510	Elective-II	3	--	2	4
PRJ-PL520	Mini Project and Seminar-I	--	--	4	2
MCC-590	Research Methodology and IPR	2	--	--	2
MCC-591	English for Research Paper Writing	2	--	--	2
	Sub-Total	16	--	14	23
Semester II					
PCC-PL510	PLM: Advanced Concepts	3	--	2	4
PCC-PL511	Web and Networking Technologies	3	--	2	4
PEC-PL511 to	Elective-III	3	--	2	4
PEC-PL519	Elective-IV	3	--	2	4
PRJ-PL521	Mini Project and Seminar-II	--	--	4	2
OEC-8**	Open Elective	3	--	--	3
AUD-9**	Audit Course	2	--	--	--
	Sub-Total	17	--	12	21
Second Year					
Semester I					
DIS-PL601	Dissertation - I	--	--	32	16
HSSM-601	Indian Knowledge System	2	--	--	02
	Sub-Total	2	--	32	18
Semester II					
DIS-PL602	Dissertation - II	--	--	36	18
CCC-602	Yoga and Sports	--	--	02	--
	Sub-Total	--	--	38	18
Grand Total		35	--	96	80

L – No. of Lecture Hours/week, T – No. of Tutorial Hours/week, P – No. of Practical Hours/week




Head
Mechanical Engineering Department
S.G.G.S.I.E. & T. Nanded.

Program Elective – I and Program Elective – II	
Course Code	Name of the course
PEC-PL504	Computer Aided Design
PEC-PL505	Computational tools and programming - I
PEC-PL506	Reliability and Life Testing
PEC-PL507	Supply Chain Management
PEC-PL508	Total Quality Management
PEC-PL509	Lean Manufacturing
PEC-PL510	New Product Design
Program Elective – III and Program Elective - IV	
Course Code	Name of the course
PEC-PL511	Customization of PLM Software
PEC-PL512	Finite Element Analysis
PEC-PL513	Computational tools and programming - II
PEC-PL514	Assembly Planning and Management
PEC-PL515	Digital Manufacturing
PEC-PL516	Customization of CAD/CAM Software
PEC-PL518	Machine Learning
PEC-PL519	Internet of Things

Open Elective Courses	
Course Code	Name of the course
OEC-801	Business Analytics
OEC-802	Industrial Safety
OEC-803	Operations Research
OEC-804	Cost Management of Engineering Projects
OEC-805	Composite Materials
OEC-806	Waste to Energy
OEC-807	Cyber Security

Audit Courses	
Course Code	Name of the course
AUD-901	Project Management
AUD-902	Disaster Management
AUD-903	Sanskrit for Technical Knowledge
AUD-904	Value Education
AUD-905	Constitution of India
AUD-906	Pedagogy Studies
AUD-907	Stress Management by Yoga
AUD-908	Personality Development through Life Enlightenment Skills

PCC-PL501 PLM FUNDAMENTALS (PLMF)

Objectives of the Course:

1. To impart the latest knowledge, principles, strategies, practices, and applications in PLM domain.
2. To provide an in-depth understanding of various applications and solutions of PLM.
3. To build conceptual foundation of PLM, along with the latest industry views on PLM applications.
4. To present frameworks which provide economic justifications for PLM projects.

Course Outcomes:

At the end of course, students will be able to;

- a) Justify importance and need of various components/elements of PLM.
- b) Able to demonstrate the benefits of PLM implementation/deployment.
- c) Distinguish various tasks terminologies used in domain of PLM.
- d) Apply and design the various strategies for process and product data management.
- e) Build /configure organizations, product structures, bill of materials, workflow, projects and requisite tasks in PLM.
- f) Evaluate and select the proper PLM system as per the need of organization.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Introduction: Overview, Need, Benefits, Concept of Product Life Cycle, Components / Elements of PLM, Emergence and Significance of PLM, PLM implementation cases in various industry verticals.

PLM Strategy and Vision: Company's PLM vision, PLM Strategy, Principles for PLM strategy, Preparing for the PLM strategy, Developing a PLM strategy, Strategy identification and selection, PLM business goals.

Information, Tools, Information systems and people involved in PLM. Product data and processes like New Product Development, Change Management, Concurrent Design & Process Management, product data linkages across the domain.

PLM Solutions: Different phases of product lifecycle and corresponding technologies, Enterprise information, knowledge and IP, Change Process, Product Structure & Configuration, Bill of Material, Requirement, Portfolio, Program & Project, Engineering Process, Supplier Relationship, Manufacturing Process, Maintenance Repair & Overhaul process and Simulation Process Management.

Human resources in product lifecycle, Methods, Techniques, Practices, Methodologies, Processes, System components in lifecycle, slicing and dicing the systems, Interfaces, Information Standards, Vendors of PLM Systems and Components.

Text Books:

- **Grieves, Michael**, Product Lifecycle Management, McGraw-Hill, 2006. ISBN 0071452303
- **Antti Saaksvuori, Anselmi Immonen**, Product Life Cycle Management - Springer, 1st Edition (Nov.5, 2003)
- **Stark, John**. Product Lifecycle Management: Paradigm for 21st Century Product Realization, Springer-Verlag, 2004. ISBN 1852338105
- **Kari Ulrich and Steven D. Eppinger**, Product Design & Development, McGraw Hill International Edns, 1999.

References:

- Relevant recent technical articles, research papers, key note addresses, etc.

PCC-PL501 PLM FUNDAMENTALS LAB (PLMF LAB)

Objectives of the course:

1. Demonstrate prerequisites of PLM
2. Make able to understand implementation procedures of PLM tools.
3. Build confidence in integrating CAD /CAE software along with PLM software.

Course outcomes:

At the end of course, students will be able to;

- a) Graduate can demonstrate practical uses of various PLM technologies.
- b) Configure the various CAD/CAE and PLM tools.
- c) Integrate the various CAD/CAE models to use in PLM applications.

Evaluation scheme:

SN	Component	Weightage (%)
1	Continuous evaluation	100

Course contents:

Introduction, Installation & maintenance of following software: Oracle / SQL Server / DB2, PLM Server, CAD Software, MS Office, Rich client, Web client, Application server, Software/ Hardware/ Network issues resolving.

CAD: Modelling (at least 5 parts) and Assembly using CAD - PLM integration.

Assembly should include top down and bottom-up approaches, Drafting (at least 1 assembly).

CAD File/data exchange amongst the various CAD software and software for CMM, CAE, CNC, CAM

FEA: Analysis (structural, thermal and both) of at least two parts, Introduction to nonlinear analysis

PLM: Exhibiting use of various modules of any PLM software through at least six assignments.

PCC-PL503 DATA MANAGEMENT (DM)

Objectives of the course:

1. Demonstrate the physical and logical database designs, database modeling, E-R model, relational, hierarchical, and network models.
2. To inculcate the use of data definition & manipulation language.
3. To pass on the concepts of distributed databases, Role of database in Enterprise systems like PDM / PLM.
4. To make aware of different issues involved in the design and implementation of database system.

Course Outcomes:

At the end of course, students will be able to;

- a) Demonstrate program-data independence, data models for database systems, database schema and database instances.
- b) Construct Structured Query Language statements used in creation and manipulation of Database.
- c) Represent the methodology of conceptual modelling through Entity Relationship model.
- d) Differentiate the methodology of logical and physical data model.
- e) Develop and evaluate a real database application using a database management system.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I - Introduction to Database Management Systems

Database-System Applications, Purpose of Database Systems, View of Data, Database Languages, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems.

Unit II - Database Design, E-R and Relational models

Overview of the Design Process, The Entity-Relationship Model, Constraints, Removing Redundant Attributes in Entity Sets, Entity-Relationship Diagrams, Reduction to Relational Schemas, Entity-Relationship Design Issues, Introduction to the Relational Model: Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages, Relational Operations, Features of good relational design, normal forms.

Unit III - SQL Concepts

Introduction to SQL: Overview of the SQL Query Language, SQL Data Definition, Basic Structure of SQL Queries, Additional Basic Operations, Set Operations, Null Values, Aggregate Functions, Nested Subqueries, Modification of the Database, Intermediate SQL, Advanced SQL.

Unit IV - Advancement in DBMS

Object-Based databases; DBMS architectures: Centralized and Client-Server Architectures, Server System Architectures, Parallel Systems, Distributed Systems, Network Types, Distributed databases: Homogeneous and Heterogeneous Databases.

Unit V - DBMS and PLM

Product Data, Product Data Management, Data Models, System Aspects, PDM Standards - STEP Standard.

Unit VI - DBMS Software

Case studies; software tools and programming using SQL; Oracle: Database, PL/SQL, developer tools; MS SQL Server, My SQL, MS Access, Visual FoxPro.

Text Books:

- Silberschatz, Korth and Sudarshan, Database System Concepts, McGraw Hill, 2002
- S. K. Singh, Database Systems: Concepts, Design and Applications, Pearson India, 2009
- Ikvinderpal Singh, Database Management Systems & Oracle, Khanna Publishing, 2014

Reference Books:

Software documentation of Oracle, MS Access, Visual FoxPro, MS SQL server, MySQL, MS Excel

- Crnkovic, Ivica; Asklund, Ulf; & Dahlqvist, Annita Persson, Implementing and Integrating Product Data Management and Software Configuration Management, Artech House Publishers, 2003. ISBN 1580534988
- Peltonen, H., Concepts and an Implementation for Product Data Management, Acta Polytechnica Scandinavica, Mathematics and Computing Series no. 105, Espoo 2000, 188 pp. Published by the Finnish Academies of Technology. ISBN 951-666-538-1. ISSN 1456-9418.

PCC-PL503 DATA MANAGEMENT LAB

Objectives:

1. To impart process of installing and using DBMS software.
2. To create tables and database in database systems.
3. To make students understand and practice use of various models like E-R model and Relational models.
4. Impart the Programming skills in DBMS packages using SQL.

Course Outcomes:

At the end of course, students will be able to;

- a) Demonstrate DBMS architecture, physical and logical database designs, database modeling, relational, hierarchical and network models and installation.
- b) Apply Structured query language (SQL) for database definition and database manipulation.
- c) Develop and evaluate a real database application using a database management system.
- d) Identify different types of database failures and techniques to recover from such failures.

Evaluation scheme:

SN	Component	Weightage (%)
1	Continuous evaluation	100

Course contents:

Study of any one DBMS Software: Oracle/ MS SQL Server/ Visual FoxPro / MS Access/ My SQL.

Queries implemented on DBMS using SQL.

Micro/Mini Project/s to be carried out throughout the semester to understand the above-mentioned concepts of Database management.

Text Books:

1. Silberschatz, Korth and Sudarshan, Database System Concepts, McGraw Hill, 2002
2. S. K. Singh, Database Systems: Concepts, Design and Applications, Pearson India, 2009
3. Ikvindepal Singh, Database Management Systems & Oracle, Khanna Publishing, 2014
4. **DBMS Software documentation.**

PEC-PL504 COMPUTER AIDED DESIGN (CAD)

Objectives of Course

1. To exhibit use of computers in design process.
2. To make aware facilities in different CAD Software
3. To inculcate theory of CAD techniques, data exchange.
4. To theorize the concepts of CAD transformation, mathematical representation of Solids, Curve and surfaces.
5. To Impart assembly modeling techniques in CAD.

Course Outcomes:

At the end of course, students will be able to;

- a) Classify CAD hardware and software for variety of applications.
- b) Exhibits importance and use of various CAD modeling techniques.
- c) Exhibits the idea and procedure of data transfer in various CAD tools.
- d) Calculate the position, shape and geometry of CAD model after translation or transformation.
- e) Demonstrate the assembly procedure of CAD applications.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Product design process: Importance of design, design process, technological innovation and the design process, Team behavior and tools; Embodiment design: Product architecture, configuration of design, parametric design, Industrial design, Human factors design, Design for X (DFX).

Unit II CAD – Introduction, Role of CAD, CAD system architecture, Hardware and software for CAD, Software modules, ICG, Graphics Software, Ground rules for design of GS, functions of GS, modeling and simulation, Solid modeling methods.

Unit III An overview of modeling software like UG/NX, Solid Works, Autodesk Inventor, Professional, AutoCAD, PRO/E, CATIA: Capabilities, Modules, Coordinate systems, Sketching tools, solid modeling tools, surface modeling tools, expression/parameters toolbox, data exchange tools, API and customization facilities.

Unit IV Geometric transformations: 2D and 3D; transformations of geometric models like translation, scaling, rotation, reflection, shear; homogeneous representations, concatenated representation; Orthographic projections.

CAD/CAM Data exchange and data storage: Introduction, graphics and computing standards, data exchange standards like IGES, STEP, Model storage - Data structures - Data base considerations - Object oriented representations - Organizing data for CIM applications - Design information system.

Unit V Mathematical representations of solids: Fundamentals, Solid models, Classification of methods of representations, half spaces, boundary representation, CSG, sweep representations, Octree representations, primitive instancing, cell decomposition, spatial occupancy enumeration.

Mathematical representations of curves and surfaces: Curve representation, parametric representation of analytic and synthetic curves; Surface models, Surface representations, parametric representation of analytic and synthetic surfaces.

Unit VI Assembly modeling: Representation, mating conditions, representation schemes, generation of assembling sequences.

Reverse engineering, evolution of AI approaches and applications in CAD, Knowledge Based Engineering, OpenGL, Introduction to Advanced visualization topics in CAD like Modern representation schemes like FBM, PM, Feature recognition, Design by features, Open-Source CAD like Open CASCADE.

Term Work:

- **Introduction, Installation & maintenance of following software:** Oracle / SQL Server / DB2, various CAD Software and Software/ Hardware/ Network issues resolutions.
- **CAD:** Modeling (at least 10 parts) and Assembly using any High-End CAD Software.
- Assembly should include top down and bottom-up approaches, Drafting (at least 3 assembly).
- CAD File/data exchange amongst the various CAD software and software for CMM, CAE, CNC, CAM.

Text Books:

- **Rogers, D.F. and Adams, A.**, Mathematical Elements for Computer Graphics, McGraw Hill Inc, NY, 1989
- **Groover M.P. and Zimmers E. W.**, CAD/CAM: Computer Aided Design and Manufacturing, Prentice Hall International, New Delhi, 1992.
- **Software Documentation, tutorials, manuals of following software namely** UG/NX, Solid Works, CATIA, Autodesk Inventor Professional, AutoCAD, Open CASCADE, ANSYS Design modeler, Pro/E

References:

- **Ibrahim Zeid**, Mastering CAD/CAM, Tata McGraw Hill Publishing Co. Ltd., New Delhi.
- **Chris McMahon and Jimmie Browne**, CAD/CAM – Principle Practice and Manufacturing Management, Addison Wesley England, Second Edition, 2000.
- **Dieter George**, Engineering Design –A materials and processing approach, McGraw Hill Publishers, 2000.
- **S. S. Pande**, Computer Graphics and Product Modeling for CAD/CAM, Narosa Publ., 2012.

PEC-PL505 COMPUTATIONAL TOOLS AND PROGRAMMING – I**Objectives:**

1. To familiarize computational tools like MATLAB/OCTAVE for Application issue solving, C, C++ programming and related Integrated Development Environments (IDEs)
2. To impart OOP (C++/Java programming) and related IDEs.
3. To Comprehend the Software Engineering.
4. Inculcate the cloud computing & IOT concepts.

Course Outcomes:

At the end of course, students will be able to,

- a) Write programs in languages like C, C++, C#, VB.
- b) Work in software development projects.
- c) Effectively use computing tools like spreadsheets, Octave for data processing, engineering calculations and write programs therein.
- d) Write algorithms and programs for development of engineering applications.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:**Unit I - Computational Tools**

Introduction to Computational Tools like MATLAB / OCTAVE, Spreadsheet.

Unit II - Programming in C Language

Problem Solving and Algorithms/Flowcharts; C Programming Fundamentals - Data Types, The Control Structure, Functions, Arrays, Pointers, Structures and File Handling, Memory Management; Integrated Development Environments (IDEs) - Editor, Compiler, Linker, Debugging, Libraries.

Unit III - Software Engineering

Introduction to Software Engineering.

Unit IV - Software Project Development in C

Algorithm and Program Development for Various Topics in Data Structures, Numerical Methods, Mechanical Design and Manufacturing, Costing and Estimation.

Unit V - OOPs and Programming in C++/ VB/ core Java

Object Oriented Programming (OOP) Concepts and Programming in C++/C#/VB/ core Java.

Term Work:

At least 10 Practical assignments on above syllabus.

Text Books:

- Y. Kanetkar, Let us C, 5th ed., New Delhi: BPB Publications, 2004, p. 746.
- E. Balagurusamy, Programming in ANSI C, 6th ed., New Delhi: McGraw Hill Education (India) Private Limited, 2015, p. 558.

Reference Books:

- Herbert Schildt, C++: The Complete Reference, Fourth Edition, McGraw-Hill Education, 2002.
- Y. Kanetkar, Let us C++, 2nd ed., New Delhi: BPB Publications, 2017, p. 633.

- Holzner Steven, Java 2 (JDK 5 Ed.) Programming Black Book, Dreamtech Publishers, 2005.
- J. Walkenbach, Microsoft Excel VBA Programming, 3rd ed., New Jersey: John Wiley and SOS Inc., 2013.
- S. C. Chapra and R. P. Canale, Numerical Methods for Engineers, 6th ed., New York: McGraw Hill Companies Inc., 2010.
- B. Maxfield, Engineering with Mathcad - Using Mathcad to create and organize your engineering calculations, Burlington: Butterworth-Heinemann publications, 2006.
- J. W. Eaton, D. Bateman, S. Hauberg and R. Wehbring, GNU Octave - A high-level interactive language for numerical computations, 4th for Octave version 4.0.3 ed., 2016.
- Roger S. Pressman, Software Engineering: A Practitioner's Approach, 7th edition, McGraw Hill Education, 2009.

PEC-PI 506 RELIABILITY ENGINEERING & LIFE TESTING (RE<)

Objectives:

1. To Demonstrate the importance of Reliability Engineering & Life Testing for manufacturing systems.
2. To impart skills for finding reliability of any component or system.
3. To inculcate how to improve the reliable life of a component or system.

Course outcomes:

At the end of course, students will be able to;

- a. Decipher multiple roots of failure that lead to Root Cause Analysis (RCA).
- b. Represent failure data statistically and to plot density function, survival probability and hazard rate by using appropriate probability distribution.
- c. To Compute system reliability using Reliability Block Diagrams.
- d. Compute reliability, MTTF and MTBF from accelerated life testing data using failure rate models.
- e. Carry out FMEA and FTA to improve machine or equipment or product reliability.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Basic concepts in Reliability: Risk and Reliability, Bathtub curve, Failure Mechanism of mechanical components: causes, modes, function of mechanical elements, failure theories. **Component Reliability:** Failure data analysis, reliability function, hazard rate, failure rate, and their relationship, MTTF, mean failure rate, MTBF.

Unit II System Reliability: Series, parallel, mixed configuration, r-out of-n structure, solving complex systems, reliability logic diagrams (RLD). Techniques of Reliability Estimation: Fault Tree analysis, tie sets and cut-sets, Boolean algebra.

Unit III System Reliability Improvement: use of better components, simplification, derating, redundancy, working environment control, maintenance, etc. Redundancy Techniques: Introduction, component vs unit redundancy, weakest link technique, mixed redundancy, standby redundancy, redundancy optimization, double failure and redundancy. **Case Application of complex systems:** Marine power plant, computer system, Nuclear power plant, combats aircraft, etc.

Unit IV Reliability Testing: Introduction, objectives, assumption, different types of test. Life testing in practice: Methodology, problems and difficulties. Economics of Reliability engineering.

Unit V Life Cycle Testing: Intro, basic concepts, data qualification. Accelerations faster, stress combination methods, limitations, step stress method for AST, various AST models, recent development recommended approach. Highly accelerated life testing (HALT), HASS

Unit VI Self-Learning Component Through Sessional: Case application, assignments, subject paper/project, presentation etc.

Term Work:

At least 8 Practical assignments on above syllabus.

Text Books:

- Srinath LS, Mechanical Reliability, Affiliated East-West Press Pvt. Ltd, New Delhi.
- Srinath LS, Reliability Engineering Third Ed., Affiliated East-West Press Pvt. Ltd, New Delhi.
- V.N.A. Naikan, Reliability Engineering and Life Testing, PHI Learning Pvt. Ltd. New Delhi.
- E. Balagurusamy, Reliability Engineering, TMH, New Delhi.

Objectives:

1. To develop an understanding of key drivers of supply chain performance and their inter-relationships with strategy and other functions of the company such as marketing, manufacturing and accounting.
2. To impart analytical and problem-solving skills necessary to develop solutions for a variety of supply chain management and design problems and develop an understanding for use of information technology in supply chain optimization.
3. To develop the ability to design logistics systems and formulate integrated supply chain strategy, so that all components are not only internally synchronized but also tuned to fit corporate strategy, competitive realities and market needs.
4. To identify improvement opportunities that exist within supply chains in different industries and to quantify the improvements that various supply chain strategies offer.

Course Outcomes:

At the end of course, students will be able to;

- a) Analyze structure of supply chains and the different ways through which supply chains can become competitive in the market.
- b) Find the method to use the levers of the logistics strategy to redefine the points necessary to make this harmonization.
- c) Analyze the importance of the term “value creation” and to propose actions in the field of management of logistics costs towards the creation of value.
- d) Computing the Demand forecasting, Aggregate Planning and Master Production Schedule, managing supply and demand.
- e) Demonstrate Use of IT services for managing Transportation and Inventory Management.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Introduction:

Definitions, Historical perspective, objectives of Supply Chain Management (SCM), Decision phases, Process Views, Strategies and achieving strategic fit, Performance measures, Drivers of SCM and associated framework, case studies.

Unit II Network Design: Role of distribution, influencing factors, design options, online sales, role of network design, influencing factors and framework for network design, models for facility location and capacity allocation, designing global supply chain networks, dealing with uncertainty, case studies.

Unit III Coordinating Demand and Supply: Demand forecasting, forecast error and its impact, role of Information Technology (IT) in forecasting, Aggregate Planning and Master Production Schedule for SCM, role of IT in aggregate planning, managing supply and demand, operational planning issues, coordination in SCM, case studies.

Unit IV Inventory Management: Cycle inventory, related costs, multi-echelon cycle inventory, managing uncertainty, role of IT in inventory management, product availability: importance, influencing factors and improvement, product availability and capacity constraints, optimal availability, case studies.

Unit V Transportation: Role of transportation, transportation: modes, performance of modes, infrastructure and policies, transportation network design, role of IT, sourcing, 3rd and 4th party logistics, supplier selection, contracts, procurement

process, pricing and revenue management: multiple customer segments, perishable assets, seasonal demand, bulk and spot contracts, case studies.

Unit VI IT in SCM: Supply chain IT framework, macro processes, case studies.

Text Books:

- **Sunil Chopra, Peter Meindl, D. V. Kalra** – Supply Chain Management: Strategy, Planning and Operations, 5th Edition, Pearson Education Inc., 2013

Reference Book

- **Rahul V. Altekhar** – Supply Chain Management: Concepts and Cases, Eastern Economy Edition, Prentice Hall of India, 2005
- **Jeremy F. Shapiro** – Modeling the Supply Chain, Duxbury, Thomson Learning Inc., 2001
- **Martin Christopher**, - Logistics and Supply Chain Management – Strategies for reducing cost and improving service, Pitman Publishing, II Edition, 1998.
- **B S Sahay**, Supply Chain Management for Global Competitiveness, Macmillan India Ltd., New Delhi, 2000.

PEC-PL508 TOTAL QUALITY MANAGEMENT (TQM)

Course Objectives:

1. To impart fundamentals of Customer satisfaction.
2. To familiarize philosophies of total quality management by renowned quality gurus.
3. To make aware of the scientific tools for quality improvement.
4. Demonstrate off-line quality control for quality improvement.
5. To inculcate the contemporary quality assurance standards.

Course Outcomes:

At the end of course, students will be able to;

- a) Compute the cost of the Quality.
- b) Implements, Perform and use Benchmarking, Brainstorming, Quality circles, for quality improvement.
- c) Select proper resources by using QFD analysis.
- d) Build confidence for better quality by using various quality control techniques.
- e) Implements the quality standards as per need of product / Organization.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course content:

Unit I Introduction: Quality revolution. The changing business conditions. Forces of competitiveness. Significance and meaning of quality. The quality functions. Various definitions of quality and their comparisons. Two-dimensional definition of quality. Eight dimensions of quality, Components of Customer satisfaction.

Quality concept of TQM, Definitions of TQM, Elements, Issues Concepts, and Principles of TQM. TQM Philosophies of Deming, Juran, P. Crosby, Imai, Ishikawa, Conway.

Unit II Assessment of Quality Cost: Objectives, Cost of poor quality, Quality cost classification, Analysis of quality cost, hidden quality costs, Economic models of quality cost, guidelines to establish and cut down quality cost.

Unit III Tools for Quality Improvement: Seven old and new Q.C. tools, Benchmarking, Quality Circles, The PDCA cycle, Hoshin Kanri Plan. Six Sigma approach.

Unit IV Quality Function Development: Concept & defining QFD, product development system, QDF process, QFD matrix concept. Deployment - part, process. T- type matrix.

Unit V Off Line Quality Control: Robust design, Loss function, Taguchi's recommended design techniques, O.A., Linear graphs, Taguchi's analysis techniques, performance measures S/N ratios, parameter design, inner and outer arrays. Design and Analysis of Experiments: Factorial experiments, Analysis of variance, Analysis of means.

Unit VI Quality Standards: ISO 9000: Concepts, methods & implementation. Quality management practices worldwide, interpretation of key ISO 9000 clauses, Implementing ISO 9000, Indian equivalent for ISO 9000, The ISO 9001:2000 standard; steps for certification under ISO9001:2000.

References Books:

1. Quality planning & analysis - J.M. Juran, Frank M. Gryna.
2. Total Quality Management – Logothetis
3. Total Quality Management – Banks
4. Fundamentals of Quality Control and Improvement – Amitava Mitra Pearson Education Inc.
5. Total Quality Control Essentials - Sarv Singh Soin - McGraw Hill Ltd.
6. Quality Circles Master Guide- Sud Ingle (PHI Publication)
7. Taguchi Techniques for quality engineering -Philip J. Ross - McGraw Hill Ltd.
8. QFD linking a company with its customers- Ronald G. Day. - McGraw Hill Ltd.
9. The complete ISO Manual - Dennis Green

Relevant recent technical articles, research papers, key note addresses, etc.

PEC- PL509 LEAN MANUFACTURING (LM)

Objective of the course:

1. To impart conceptual understanding of JIT Logic along with Pull and Push production system.
2. To impart the importance of Implementation of JIT principles to waste elimination along with understanding of Japanese approaches.
3. To emphasis on Kanban system to counter problems & dealings of both suppliers and contractors with the help of related Kanban cards.
4. To make understanding of the rise of lean production along with birthplace, concrete example, company as community, final assembly plant, product development and engineering, changing customer demand and future of lean production.
5. To promote people for creating an organization and installing business system to encourage lean thinking.

Course Outcomes:

At the end of course, students will be able to;

- a) Recognize the importance of Just In Time Production System.
- b) Analyze and evaluate problems related to Kanban system.
- c) Apply lean manufacturing tools to reduce lead time.
- d) Use elements of lean production to manage lean enterprise.
- e) Understand the background behind the rise of lean production.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Introduction to Lean Manufacturing: Production System and its types, Transition to Lean, Lean Thinking, Manufacturing Strategies, Benefits of Lean Manufacturing.

Unit II Elements of Lean Manufacturing: Elimination of Waste, Value Stream Mapping, 5S, Kaizen Approach, Introduction to and comparative study of Toyota Production System, Total Productive Maintenance, Total Quality Management and Six Sigma, Lean Indicators and Organizational Performance.

Unit III Cellular Manufacturing: Layouts, Group Technology - part families, clustering methods - Rank Order Clustering, Single-Pass Heuristic considering Capacities (Askin and Standridge), Clustering using Similarity Coefficients, Production Flow Analysis, Utility Layout.

Unit IV Just In Time Production System: JIT Philosophy, JIT implementation requirements, Production Smoothening – philosophy and methods, Pull system - Production Authorization, Kanban Systems, scheduling Kanban production, CONWIP system, Base Stock System, Inventory Management in JIT, Information Management in JIT.

Unit V Shortening of Production Lead Times: Reduction of setup times, practical procedures for reducing setup time, Transfer Lots, Economic implications of setup time reduction, Standardization of operations, multi function workers and job rotation.

Unit VI Human Approach for Lean Implementation: Lean Leadership, Total Employment Involvement, Small Group Activities like Quality Circles, SMTs, etc.**Scheduling:** Scheduling System Requirements, Bottleneck Scheduling, Single Machine Scheduling, Flow Shop Scheduling, Job Shop Scheduling.

Term Work:

At least 08 Practical assignments on above syllabus.

Text Books

- **Ronald G. Askin and Jeffrey B. Goldberg**, “Design and Analysis of Lean Production Systems”
- **Chasel Aquilino**, “Productions and Operations Management”

Reference Books:

- **Yasuhiro Monden**, “Toyota Production System -An integrated approach to Just in Time”, Engineering and Management Press, Institute of Industrial Engineers, Norcross Georgia
- **James P Womack, Daniel T Jones, and Daniel Roos**, “The Machine that changed the World. The Story of Lean Production”, Harper Perennial edition, 1991.
- **James Womack**, “Lean Thinking”.
- **Richard Schourberger**, “Japanese Manufacturing Techniques. The Nine Hidden Lessons by simplicity”.
- **Jeffrey Liker** “The Toyota Way : 14 Management Principles from the World's Greatest Manufacturer: 14 Management Principles from the World's Greatest Manufacturer”

PEC-PL510 NEW PRODUCT DESIGN (NPD)

Objective of the course:

1. To provide conceptual understanding of product design, product development process & methodologies.
2. To integrate product development process by identifying customer needs by gathering, interpreting, organizing, and establishing relative importance of the customer needs.
3. To highlight on complete design, justification and analysis (simulation), tool design, plan manufacturing, material and process selection, tools and software selection, testing (quality check) and servicing the product.
4. To promote students for selecting and solving cases from various sectors with the help of product and process systemization, identification and solving methodologies, improving product development solutions.

Course Outcomes:

At the end of course, students will be able to;

- a) Identify and analyses the Product Design and Development processes in manufacturing industry.
- b) Analyze and carry out PMI, FMEA, SWOT analysis for new product to be designed.
- c) Perform the QFD, KANO Model, and Risk analysis for proper selection of elements to be considered for new product design.
- d) Determine the time and cost required for new product design through various business analysis tools.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Engineering Design Process: Problem solving methodology for engineering design, design considerations, design process, design review.

Unit II Product Development Process and Need Assessment: Introduction, Product Development Process, Product and Process cycle, Technological Innovation, Identifying customer need, Engineering Characteristics, Product Design specification.

Unit III Idea Generation and Screening: Tools and Solution for idea generation are Brainstorming, Competitive Intelligence, Conjoint Analysis, Delphi Technique, Morphological Charts, Six Thinking Hats, TRIZ-idea generation using problem solving tool, Tools and Solution for Idea Screening are Qualitative Research, FMEA, SWOT Analysis, PMI Analysis.

Unit IV Concept Development and Testing: Controlled Convergence, Risk Management, Concept Testing like Quantitative Research, Quality Function Deployment, KANO Model.

Unit V Business Analysis and BETA: Product Life Cycle, Gantt Charts, Critical Path Analysis and PERT, Rapid Prototyping. **Technical Implementation:** Design for X, Industrial Design, Re-engineering, Reverse Engineering.

Unit VI Product Consideration and General-Purpose Tools: Marketing Plan, Distribution Channels, Product Pricing, Benchmarking Tool, Business Process Re-engineering, Product outsourcing.

Text Books:

- Dieter George E., Engineering Design, McGraw Hill Pub. Company, 2000.

References:

- Ulrich Karl T and Eppinger Steven D., Product design and development, McGraw Hill Pub. Company, 1995.
- Chitale A. K. and Gupta R. C., Product Design and Manufacturing, Prentice-Hall of India, New Delhi, sixth edition.
- Bralla, James G., Handbook of Product Design for Manufacturing, McGraw Hill Pub. 1986
- Relevant recent technical articles, research papers, key note addresses, etc

PRJ-PL520 MINI PROJECT & SEMINAR-I

Objectives of the course:

1. To motivate for using recent technical research papers and articles.
2. To promote for improving learning, presenting and documentation skills.

Course Outcomes:

- a. Apply effective strategies in literature searches using available resources.
- b. Build programs, software tool, interface for automating PLM system issues.
- c. Document properly according to a specific style.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course contents:

The Seminar and Mini Project shall consist of few particulars amongst **literature review** based on a sizable number of publications. **Design / Development / Synthesis** related to a particular area. Implementation of existing theory for applications, pilot experiments etc. Each student is required to prepare a report and deliver a talk based on the work carried out as mini-project under the guidance of a faculty member(s). The work carried out should be preferable related to his/her dissertation topic.

MCC-590 RESEARCH METHODOLOGY AND IPR (RMIPR)**Objectives of the course:**

1. Understand research problem formulation.
2. Analyze research related information.
3. Follow research ethics.
4. Understand that today's world is controlled by Computer, Information Technology, but tomorrow world will be ruled by ideas, concept, and creativity.
5. Understanding that when IPR would take such important place in growth of individuals and nation, it is needless to emphasis the need of information about Intellectual Property Right to be promoted among students in general and engineering in particular.
6. Understand that IPR protection provides an incentive to inventors for further research work and investment in R and D, which leads to creation of new and better products, and in turn brings about, economic growth and social benefits.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course Contents:

Unit 1: Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations

Unit 2: Effective literature studies approaches, analysis Plagiarism, Research ethics,

Unit 3: Effective technical writing, how to write report, Paper

Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee

Unit 4: Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.

Unit 5: Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.

Unit 6: New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.

References:

1. Stuart Melville and Wayne Goddard, "Research methodology: an introduction for science & engineering students"
2. Wayne Goddard and Stuart Melville, "Research Methodology: An Introduction"
3. Ranjit Kumar, 2 ndEdition , "Research Methodology: A Step by Step Guide for beginners"
4. Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd ,2007.
5. Mayall , "Industrial Design", McGraw Hill, 1992.

6. Niebel , “Product Design”, McGraw Hill, 1974.
7. Asimov , “Introduction to Design”, Prentice Hall, 1962.
8. Robert P. Merges, Peter S. Menell, Mark A. Lemley, “ Intellectual Property in New Technological Age”, 2016.
9. T. Ramappa, “Intellectual Property Rights Under WTO”, S. Chand, 2008

Objectives of the course:

1. Understand that how to improve your writing skills and level of readability.
2. Learn about what to write in each section.
3. Understand the skills needed when writing a Title.
4. Ensure the good quality of paper at very first-time submission.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course Content:

Unit 1: Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness

Unit 2: Clarifying Who Did What, Highlighting Your Findings, Hedging and Criticising, Paraphrasing and Plagiarism, Sections of a Paper, Abstracts. Introduction

Unit 3: Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check.

Unit 4: Key skills are needed when writing a Title, key skills are needed when writing an Abstract, key skills are needed when writing an Introduction, skills needed when writing a Review of the Literature,

Unit 5: Skills are needed when writing the Methods, skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions

Unit 6: Useful phrases, how to ensure paper is as good as it could possibly be the first- time submission

References:

1. Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)
2. Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press
3. Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman's book .
4. Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011 .

PCC-PL510 PLM: ADVANCED CONCEPT (PLM-AC)

Objectives of the course:

1. To demonstrate the important concepts in PLM like Multisite collaboration, change management, Legacy System Integration & data transfer, product architecture, and CAD BOM alignment.
2. To impart the use of workflow, product structuring, visualization of data, and architectures of PLM systems.
3. To comprehend various elements of PLM like Security Management, Item management, Document management and Configuration Management.
4. To promote for using, learning and demonstrating new trends in PLM.
5. To motivate to work in multi cad and environment.

Course outcomes:

At the end of course, students will be able to;

- a) Graduate exhibits the use of various elements of PLM.
- b) Selects proper architecture of PLM tools for an Industry.
- c) Administer for installing, configuring, integrating and efficient management of PLM tools.
- d) Select proper methodology for data transfer and implementation in CAD and PLM systems.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I: Multisite, how to configure Multisite? Importing other sites to FMS MASTER, Replica Creation (Cloning DB for test or QA), Data share and import-export between multisite, Global Change Management System (GTS), Legacy System Integration, Legacy Data Transfer, Security in PLM (SSO/SSL etc),

Unit II: Product master management (managing the deployment of the finished design into the production environment), product architecture (Functional architecture, Physical architecture etc), understanding business object, CAD-BOM alignment, security services, PLM localization, Business modeling, classification structure, PLM System Architecture (2tier/3tier/4tier etc) Managing Changes and Workflows, Classifying Data, Managing Documents, Reports, Requirements, and Schedules, Sharing Data, Managing Product Structures, Managing Manufacturing Data, Visualizing Products, Managing CAE Data, Repeatable Digital Validation, Managing Quality Data, Managing Maintenance, Repair, and Overhaul Data.

Unit III: Product Data: Data objects to represent product data, such as parts, assemblies, processes, product changes, requirements, and specifications, Simple parts (with JT /with CAD /with CAD+JT/ with CAD + drawing / with CAD + JT + drawing + other documents), Simple assembly, multilevel assembly, Hybrid assembly, concurrency in data transfer (replica transfer/delta transfer/re-export), collision.

Unit IV: Concepts of Product Structure management such as Configurations, Multi CAD Integrations, issues involved, data management of heterogeneous CAD systems, management of product data interfaces, GD&T, annotations, manufacturing notes, Integration of CAM with PLM. Introduction to augmented reality, Digital twin and IOT

Unit V: PLM Implementation: Activities Involved under various phases of PLM implementation like Pre-Align, Align, Plan, Build, Test, Deploy and Close. Project Planning, Documentation and Deliverables involved.

Text Books:

- **Grieves, Michael**, Product Lifecycle Management, McGraw-Hill, 2006. ISBN 0071452303
- **Antti Saaksvuori, Anselmi Immonen**, Product Life Cycle Management - Springer, 1st Edition (Nov.5, 2003)
- **Stark, John**. Product Lifecycle Management: Paradigm for 21st Century Product Realization, Springer-Verlag, 2004. ISBN 1852338105
- **Kari Ulrich and Steven D. Eppinger**, Product Design & Development, McGraw Hill International Edns, 1999.

References: Relevant recent technical articles, research papers, key note addresses, etc.

PCC-PL510 PLM: ADVANCED CONCEPT (PLM-AC) LAB

Objectives of the course:

1. Demonstrate various functions of the PDM and PLM tools.
2. Provide opportunity to enhance skills needed in domain of PLM.
3. Impart use of various applications of CAD and PLM tools.

Course outcomes:

At the end of course, students will be able to;

- a) Graduate exhibits the skill of using advanced Product lifecycle Management Tools.
- b) Selects proper architecture of PLM tools for an Industry.
- c) Integrate various CAD /data management tools with PLM systems.
- d) Administer for installing, configuring, integrating and efficient management of PLM tools.
- e) Investigate and solve various problems, errors in use of PLM tools.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Continuous evaluation	100

Course contents:

PDM Functions - Workflow Management, Project Management, Search Management,

Product Lifecycle Management (PLM) Concept and Special Functions - Creating Organization (Users, Roles, Group, Volume etc), Defining rights (Object/Rule Based), Creating required hierarchy of folders, item, form, dataset types, Defining business model Customizing different queries and reports out of the box, Creating different workflows, Creating and managing engineering change, Adding custom attribute to forms / in class, Creating different BOM view (PSE), Resource classification.

CAD Integration - CAD Manager/ Embedded Client, Seed/Template Creation, Attribute Mappings – NX/AutoCAD/ Solid Edge/Creo.

Sample Data Migration

Testing & QA.

PCC-PL511 WEB AND NETWORKING TECHNOLOGIES (W&NT)

Objectives of the course:

1. To impart the importance web engineering skills.
2. Provide opportunity to enhance skills in networking, web programming.
3. To inculcate the web developing tools like XML, HTML, and J2EE technologies.

Course Outcomes:

At the end of course, students will be able to;

- a) Select proper web and internet network.
- b) Gain knowledge of Networking Protocols, Internet working concepts, devices, Internet Basics, History and Architecture
- c) Understand working of WWW, HTTP, TELNET, JavaScript and Ajax, PHP/MySQL, ASP.Net.
- d) Gain knowledge of Java Web Technologies.
- e) Select proper Web Security and Network Security.
- f) Understand the concept of XML and Cloud Computing.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit-I Web and Computer Networks.

Unit-II Networking Protocols, Internet working concepts, devices, Internet Basics, History and Architecture.

Unit-III TCP-IP Part IV (WWW, HTTP, TELNET), JavaScript and Ajax, PHP/MySQL-An overview, ASP.Net-An Overview.

Unit-IV Java Web Technologies

UNIT-V Web Security and Network Security.

Unit-VI Introduction to XML and Cloud Computing.

Practical/ Lab work

At least six Practical assignments on above syllabus.

Text Books:

- Achyut S. Godbole and Atul Kahate, 'Web Technologies' Edition: 3
- Raj Kamal, 'Internet & Web Technologies'

Reference Book:

- Jennifer Niederst, Learning Web Design 2nd Edition
- Elizabeth Castro, HTML for the World Wide Web
- Rod Johnson, Expert One-on-One J2EE Design and Development
- Pressman and Lowe, Web Engineering-A Practitioner approach, TMH, 2008.

PEC-PL511 CUSTOMIZATION OF PLM SOFTWARE (CPLMS)

Objectives of the Course:

1. Demonstrate rapid development concepts, SDLC and Prototyping.
2. Add more features and functions in the existing PLM software tools.
3. To make aware of user interface customization, e.g., Icon/ menu, naming and arrangement.
4. Develop PLM Software customization skills.

Course outcomes:

At the end of course, students will be able to;

- a) Identify the correct customization tools for implementing PLM in Organization /Firm.
- b) Customize PLM Software as per the requirement of Organization /Firm.
- c) Suggest proper methodology for implementation of PLM through application data modelling.
- d) Customize the PLM system for server-side applications.
- e) Customize the Rich client applications for PLM system as per the requirement of Organization /Firm.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course Contents:

Unit I: Introduction to customization, need, types; introduction, Basic customization concepts, common customization tasks, software engineering concept; Software Development Life Cycle (SDLC), Requirement analysis, Rapid application Development (RAD) tools, programming languages. Customization of Word processing and spreadsheet tools.

Unit II: PLM Software Data & Application Modeling: Data and Application Modeling Interface like View, Perspective, Menu usage, connection, Overview, Codeless and Codeful customization of Data and Application Modeling.

Unit III: Server-Side Customization: Understanding Server-Side API, Finding APIs for your needs, writing new APIs, Samples, Call Server-Side Program/Code from the Client-Side Environment.

Unit IV: Client-Side Customization: Client-Side Platform, Client Side Non programming customization Client-Side customization environment setup, Basics Client-Side Plug-in, Overriding Command, Adding menu command to a menu, toolbar and shortcut menu; Client-Side-UI form introduction.

Unit V: Service Based Customization: Basic of Service based framework, Using Existing Service based API, writing new Service API, Calling API through rich client and thin client.

References:

- a) PLM Software Customization Documentation/Help Manuals,
- b) Getting started with Customization,
- c) Client-Side Customization- Programmer's Guide,
- d) Integration Toolkit Programmer's Guide
- e) Data & Application Modeling Guide,
- f) Application interface Web Service (AIWS) configuration and customization Guide.

PEC-PL 511 CUSTOMIZATION OF PLM SOFTWARE LAB (CPLM LAB)

Objectives of the Course:

1. Demonstrate PLM Software Application and Data Model Administration
2. Provide opportunity to enhance skills of PLM Software Installation and Integration
3. To inculcate PLM Software Server and Client Customization

Course outcomes:

At the end of course, students will be able to;

- a) Create custom Item, Item Revision, Form, Rules, LOVs
- b) Customized the Server as per the requirement of the Organization / Industry.
- c) Writing the new Application Program Interfaces for retrieving and uploading data from the server.
- d) Writing new Services APIs for integrating third party applications.
- e) Use the out of the box (OOTB) utilities for customizing the Server, Clients, and Data models as per the requirement of the Organization / Industry.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Continuous evaluation	100

Course Content:

It will consist of Assignments on Following Topics

- PLM Software Data & Application Modeling (creating Items, Forms, LOVs, Options, Extensions, Different Rule like Naming, GRM, Deep Copy etc); Codeless and Codeful customization.
- Server-side Customization for creating Items, Objects, Custom Handlers, writing new APIs, Samples, Call server function from the Rich Client.
- Client-side Customization for creating User Interface; Adding Custom menu bar, custom menu items, using Eclipse Rich Client Plug - in.
- Service Based Customization for Writing new Services API; Calling API through rich client and thin client.

References:

- a) PLM Software Customization Documentation /Help Manuals,
- b) Getting started with Customization,
- c) Client-Side Customization- Programmer's Guide,
- d) Integration Toolkit Programmer's Guide
- e) Data & Application Modeling Guide,
- f) Application interface Web Service (AIWS) configuration and customization Guide.

PEC-PL512 FINITE ELEMENT ANALYSIS (FEA)

Course Objectives:

1. To make aware of the general steps of finite element methods and basic finite element formulation techniques.
2. To theorize the challenges Finite Element Modeling.
3. To exhibit the importance Finite Element Techniques.

Course Outcomes:

At the end of course, students will be able to;

- a) Derive equations in finite element methods for 1D, 2D and 3D problems.
- b) Formulate and solve basic problems in heat transfer, solid mechanics and fluid mechanics.
- c) Write computer program based on finite element methods.
- d) Use of a commercial software, to solve basic engineering problems in heat transfer, solid mechanics and fluid mechanics.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I: Introduction to Finite Element Method: Basic Concept, Historical Background, Engineering applications, general Description, comparison with other methods.

Unit II: Finite Element Modeling: Introduction, Mesh Generation, mesh requirements, Semi-Automatic Methods, Node-based approach, Region based approach, Solid-modeling-based methods. Fully Automatic Methods- Element-based approach, Application. Modeling Hints - utilizing symmetry – symmetric and antisymmetric B.C, proper and effective usage of different types of elements, warping limit, corner angle, aspect ratio, Acceptable and Unacceptable Distortion, Mesh Refinements using Isoparametric Finite Elements, Meshing in high gradient areas, Transition Regions. Sub modeling Concept, Interface of CAD and FEA software packages.

Unit III: Finite Element Techniques: Applications to solid and structural mechanics problems: External and internal equilibrium equations, one-dimensional stress-strain relations, plane stress and strain problems, axis symmetric and three-dimensional stress strain problems, strain displacement relations, boundary conditions compatibility equations, analysis of trusses, frames and solid of revolution, computer programs.

Applications to heat transfer problems: Variational approach, Galerkin approach, one dimensional and two-dimensional steady state problems for conduction, convection and radiation, transient problems.

Unit IV: Parameters affecting Accuracy of the FEA results: How to validate and check accuracy of FEA results? Computational accuracy: strain energy norm, residuals, Reaction forces and moments; convergence test, Average and unaverage stress difference. Correlation with actual testing: strain gauging-stress comparison; natural frequency comparison; Dynamic response comparison, temperature and pressure distribution comparison.

Practical/ Lab work

At least six Practical assignments on above syllabus.

Text Books:

- **R.D. Cook, D. S. Malku**, Concepts and applications of Finite Element Analysis, John Wiley and Sons, New York, Second Edition, 1981
- **J. N. Reddy**, An introduction to Finite Element Analysis, Tata McGraw- Hill Pub. Co., 2005.

Reference Books:

- **T. J. R. Hughes**, The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, Dover Publications, 2000
- **Chandrupatla and Belegundu**, Introduction to Finite Elements in Engineering. Prentice Hall India, 2003

PEC-PL513 COMPUTATIONAL TOOLS AND PROGRAMMING –II (CT&P-II)**Objectives:**

1. Develop Advance Computational tools like MATLAB/OCTAVE for Application issue solving, C, C++ programming and related Integrated Development Environments (IDEs).
2. Impart Advance Java programming and related IDEs.
3. Inculcate the process of the Advance Software Development.

Course Outcomes:

At the end of course, students will be able to;

- a) Construct the various programs for development of software.
- b) Predict the output of various programs.
- c) Build various interfaces through programming skill sets.
- d) Solve real life issues by using various computational tools and programming languages.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

- Advance concepts in Spreadsheet
- Advance Programming in MATLAB/OCTAVE/PTC MathCAD.
- Introduction to .Net
- OOPS using C++, C#
- Advance Java Programming
- Software development using PL/SQL
- DBMS Software development

Practical/ Lab work:

At least six Practical assignments on above syllabus.

Text Books / Documentation:

1. **Holzner Steven**, Java 2 Programming Black Book, Dreamtech Publishers
2. **Savitch**, Java Programming
3. **Yashwant Kanetkar**, Visual C++ Programming, BPB Publications, 1998
4. **Herbert Schildt**, OOP with C++

Reference Books:

1. Date C J, "An Introduction To Database System", Addison Wesley
2. Korth, Silbertz, Sudarshan, "Database Concepts", McGraw Hill
3. Elmasri, Navathe, "Fundamentals Of Database Systems", Addison Wesley
4. Leon and Leon, "Database Management System", Vikas Publishing House.
5. DBMS Software documentation,

PEC-PL514 ASSEMBLY PLANNING AND MANAGEMENT (APM)

Objectives of the course:

1. To create interest in the assembly line design practices prevalent in industry.
2. To gain ability to recognize situations in an assembly system environment those suggest the use of certain quantitative methods to assist in decision making.
3. To analyze and solve assembly system problems using people, skills (predominantly) and technology.
4. To increase knowledge and broaden perspective of the "industrial world" in which one will contribute his / her talent and leadership as an Industrial Engineer.

Course outcomes:

At the end of course, students will be able to;

- a) Determine the requisite layout of the manufacturing system as per industry needs.
- b) Calculate number of workstations, cycle time, smoothness index to improve the performance of assembly line.
- c) Suggest a proper assembly line design methodology to manufacturing or service industry.
- d) Apply various optimization techniques to improve the performance of assembly lines.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course Contents:

Unit I: Introduction: Assembling a product, manual and automatic assembly, robotic assembly, Liaison diagram, assembly process, key characteristics of assembly, variation risk and its management.

Unit II: Assembly Sequence Planning: Introduction, assembly sequence design process, Bourjault method of generating all feasible sequences, cutest method, stability of subassemblies, software.

Unit III: Assembly Line Design: Process of Assembly Line Design (ALD), components of ALD, consideration of equipment's, buffers, etc. Introduction to assembly line balancing and defining assembly line balancing problem using precedence diagrams.

Unit IV: Simple Assembly Line Balancing Problem (SALBP): Performance Characteristics, types of SALBP, optimal solution methods for SALBP, heuristics and meta-heuristics, introduction to Genetic Algorithm, applying simple genetic algorithmic approach to SALBP.

Unit V: Generalized Assembly Line Balancing Problem (GALBP): Considerations leading to GALBP, formulation and solution approaches for a few types of GALBP such as assignment restrictions, mixed model ALBP, U-line ALBP, parallelization, etc.

Unit VI: Reconfiguration: Need and importance of reconfiguration / rebalancing, approaches for reconfiguration.

Practical/ Lab work:

At least six Practical assignments on above syllabus.

Text Books:

- Daniel E. Whitney, Mechanical Assemblies, Oxford University Press, 2004

References:

- Mikell P. Groover, Automation, Production Systems and Computer Integrated Manufacturing, Second edition, Prentice Hall of India, 2002
- **Relevant Research Papers such as and not limited to:**
 - a. Baybars, I., 1986, A survey of exact algorithms for the simple assembly line balancing problem, Management Science 32, 909-932.
 - b. Becker, C., Scholl, A., 2006, A survey on problems and methods in generalized assembly line balancing, European Journal of Operational Research 168, 694 - 715.
 - c. Falkenauer, E., 2005, Line balancing in the real world. In: Proceedings of the International Conference on Product Lifecycle Management PLM 05, Lumiere University of Lyon, France, 2005 (http://www.optimaldesign.com/Download/OptiLine/Falkenauer_PLM05.pdf)

- Gökcen, H., Erel, E., 1998, Binary integer formulation for mixed-model assembly line balancing problem, *Computers & Industrial Engineering*, 34, 451-461.
- Gonçalves, J.F., Almeida, J.R., 2002, A hybrid genetic algorithm for assembly line balancing, *Journal of Heuristics* 8, 629-642.
- Hackman, S.T., Magazine, M.J., Wee, T.S., 1989, Fast, effective algorithms for simple assembly line balancing problems, *Operations Research* 37, 916-924.
- Helgeson, W., and Birnie, D., 1961, Assembly Line Balancing Using the Ranked Positional Weight Technique, *Journal of Industrial Engineering* 12, 394-398.
- Malakooti, B., 1994, Assembly line balancing with buffers by multiple criteria optimization, *International Journal of Production Research* 32, 2159-2178.
- Merengo, C., Nava, F., Pozetti, A., 1999, Balancing and sequencing manual mixed-model assembly lines. *International Journal of Production Research* 37, 2835-2860.
- Scholl, A., Becker, C., 2006, State-of-the-art exact and heuristic solution procedures for simple assembly line balancing, *EJOR*, 168, 666 - 693.
- N. Boysen, M. Fliedner, and A. Scholl, "A classification of assembly line balancing problems," *Jenaer Schriften zur Wirtschaftswissenschaft*, 12/06, University of Jena, 2006a.
- N. Boysen, M. Fliedner, and A. Scholl, "Assembly line balancing: Which model to use when?," *Jenaer Schriften zur Wirtschaftswissenschaft*, 23/06, University of Jena, 2006

PEC-PL515 DIGITAL MANUFACTURING (DM)

Objectives:

1. To theorize the challenges faced by manufacturing.
2. To exhibit the importance of digital manufacturing for business processes.
3. To demonstrate the importance of DM in Product Lifecycle Management.
4. To make aware of developing the digital work environment.
5. To make competent in using computer-aided technology to support the above.

Course outcomes:

At the end of course, students will be able to;

- a. Graduate exhibits the use of various elements of digital manufacturing tools.
- b. Selects proper procedure of validating practical work through digital validation.
- c. Administer for installing, configuring, integrating digital manufacturing tools.
- d. Analyze and optimize various practical manufacturing processes through digital simulation.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I: Introduction to Digital Manufacturing: A Brief History of Manufacturing, Digital Manufacturing Today, Digital Design, Digital Materials, Digital Fabrication, Digital Products, Technology Development, Applications Development, People and Business, The Digital Economy, Transition from Industrial Manufacturing.

Unit II: Process simulation and validation: Assembly and component manufacturing, process simulation and validation, Ergonomic/ human simulation, Robotic simulation and OLP.

Unit III: Plant design, simulation & optimization: Station / work-cell layout design, Throughput simulation, Discrete event simulation, Optimization of material flow and logistic.

Unit IV: Manufacturing process simulation solution customization: Functionality enhancements as extensions of OOTB software solution, Reports customization, User interface customization.

Unit V: Special Topics: Informatics platform for designing and deploying e-manufacturing systems, framework for integrated design of Mechatronic systems, Collaborative supplier integration for product design and development. Reconfigurable manufacturing systems design, Virtual Reality based platform for collaborative product review and customization, managing collaborative process planning activities through extended enterprise, rapid product development, desktop assembly factories, Information sharing in digital manufacturing based on STEP and XML

Practical/ Lab work

At least six Practical assignments on above syllabus.

Text Books:

- Wang, Lihui; Nee, Andrew Y.C. (Eds.) Collaborative Design and Planning for Digital Manufacturing, Springer, 2009.

References:

- Mikell P.Groover, Automation , Production Systems and Computer Integrated Manufacturing, Second edition, Prentice Hall of India, 2002.
- Bangsow Steffen, Manufacturing Simulation with Plant Simulation and SimTalk, Springer, 2010.
- Bangsow Steffen, Technomatic Plant Simulation-Modeling and Programming by means of examples, 2020.
- Zhuming Bi, Practical Guide to Digital Manufacturing, 2021.
- Roberta S. Russell and Bernard Taylor III, Operations Management, Wiley, Seventh Edition.
- Relevant recent technical articles, research papers, key note addresses, etc.

PEC-PL516 CUSTOMIZATION OF CAD/CAM SOFTWARE (CCCS)

Objectives:

1. Add more features and function to the existing command, CAD template designing, CAE analysis template customization.
2. To make aware Interface customization, e. g icon/menu, naming and arrangement.
3. Comprehend AutoCAD/CATIA and VBA customization techniques.
4. Inculcate the API in UG/NX and AutoCAD.

Course outcomes:

At the end of course, students will be able to,

- a. Identify custom software development requirements related to CAD/CAM applications.
- b. Use basic customization tools including macros, journals, script programs, IDEs for APIs.
- c. Customize line-types, hatch patterns, commands, shortcuts, etc in CAD/CAM software.
- d. Customize CAD/CAM Software UIs including menu, ribbon, toolbar interfaces.
- e. Design and develop user friendly and complete UIs for custom tool development in CAD/CAM software.
- f. Design and develop algorithms and programs for modeling parts for custom tool development in CAD/CAM software.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course Content:

Unit I - Introduction To CCCS

What is Customization, Need of CAD/CAM Customization, History, Part Library and Part Family in CAD, User Defined Function (UDF), Creation of Custom Templates, Macros, Linetypes and Hatching Patterns, Script Programming.

Unit II - UI Customization for CCCS

AutoCAD and NX UI Customization.

Unit III - Introduction to Programming for CCCS

IDEs, APIs in CAD Software, General Purpose and System Dependent API, Projects - Sample Projects, Objects, Methods and Properties, Data types.

Unit IV - Programming for Modeling (2D/3D)

Creating 2D and 3D entities through Programming; Editing 2D and 3D entities through Programming.

Unit V - UI Design / Development

Forms Design, Events, User Interactions and Utility objects, Controlling Menus, Toolbars, Layers and Linetypes.

Unit VI: Additional Topics in Programming for CCCS

Dimensions, Annotations, Layout, Creating Tables, File Operations, CAM Applications development using DBMS and SQL Programming.

Practical/ Lab work:

At least six Practical assignments on above syllabus.

Text Books / Documentation:

- Autodesk AutoCAD VBA Developer Documentation Help.
- GRIP programming manuals for Unigraphics (Vol. 1 and 2).

Reference Books / Documentation:

- Joe Sutphin, AutoCAD 2006 VBA: A Programmer's Reference, Apress; 2nd ed. edition (September 23, 2005).
- Marion Cottingham, Mastering AutoCAD VBA, Sybex; 1 edition (March 16, 2001).
- Lee Ambrosius, AutoCAD Platform Customization: User Interface, AutoLISP, VBA, and Beyond, Sybex; 1 edition (April 27, 2015).
- Sham Tickoo, Customizing AutoCAD 2020, 13th Edition, CADCIM Technologies (September 11, 2019).
- User Function Programming manuals for Unigraphics (Vol. 1, 2, 3).

PEC-PL518 Machine Learning (ML)

Objectives:

1. To introduce students to the basic concepts and techniques of Machine Learning.
2. To develop skills of using recent machine learning software for solving practical problems.

Course outcomes:

At the end of course, students will be able to,

- a. Appreciate the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
- b. Understand the concept behind neural networks for learning non-linear functions.
- c. Design and implement various machine learning algorithms in a range of real-world applications.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course Content:

Unit I - Introduction: History of Machine Learning, Programs vs learning algorithms, Machine Learning definition, Components of a learning, Different Types of Learning.

Unit 2: Supervised Learning (Regression/Classification): Distance-based methods, Nearest Neighbours, Decision Trees, Nave Bayes, Linear models such as Linear Regression, Logistic Regression, Generalized Linear Mode, Support Vector Machines, Nonlinearity and Kernel Methods, Multi-class/Structured Outputs, Ranking.

Unit 3: Unsupervised Learning: Clustering using K-means/Kernel K-means, Dimensionality Reduction using PCA and kernel PCA, Matrix Factorization and Matrix Completion, Generative Models such as mixture models and latent factor models.

Unit 4: Evaluating Machine Learning algorithms and Model Selection, Introduction to Statistical Learning Theory, Ensemble Methods like Boosting, Bagging, and Random Forests.

Unit 5: Sparse Modeling and Estimation, Modeling Sequence/Time-Series Data, Deep Learning and Feature Representation Learning.

Unit 6: Scalable Machine Learning (Online and Distributed Learning)- A selection from some other advanced topics, e.g., Semi-supervised Learning, Active Learning, Reinforcement Learning, Inference in Graphical Models, Introduction to Bayesian Learning and Inference.

Practical/ Lab work:

At least six Practical assignments on above syllabus.

Text Books:

- Kevin Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.
- Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning", Springer 2009.

Reference Books:

- Christopher Bishop, "Pattern Recognition and Machine Learning", Springer, 2007.
- Goodfellow, I., Bengio, Y., and Courville, A., "Deep Learning", MIT Press, 2016.

PEC-PL519 Internet of Things (IoT)

Objectives:

1. To introduce the concept of the Internet of Things and its horizon in terms of application.
2. To make aware the importance of sensor network and interoperability, and open-source physical computing platform Arduino.
3. To familiarise the programming languages such as Python and Raspberry Pi and extends its application to software -defined networking.
4. To explore the relationship between IoT, cloud computing, and big data.
5. To understand the IoT in the context of PLM.

Course outcomes:

At the end of course, students will be able to,

- a. Appreciate the potential of IoT and a lot better versed in the various things that likely to comprise the Internet of Things.
- b. Exhibit the fundamental programming skill essential for IoT implementation.
- c. Integrate the PLM and IoT platform for developing the smart tool to maximize output.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course Content:

Unit I: Introduction [7 hrs]

Introduction to IoT, Sensing, Actuation, Basics of Networking, Communication Protocols.

Unit II: Sensor Networks and Interoperability in IoT [10 hrs]

Sensor Networks, Machine-to-Machine Communications, Interoperability in IoT, Introduction to Arduino Programming, Integration of Sensors and Actuators with Arduino.

Unit III: Introduction to Python programming, Raspberry Pi and Software Defined Network [10 hrs]

Introduction to Python programming, Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi, Introduction to Software Defined Network, Software Defined Network for IoT.

Unit IV: Data Handling and Analytics and Cloud Computing [8 hrs]

Introduction to Data Handling and Analytics, Cloud Computing, Sensor-Cloud, Fog Computing, Smart Cities and Smart Homes, Connected Vehicles, Smart Grid.

Unit V: Industrial IoT and PLM [10 hrs]

Case studies on Manufacturing, Healthcare, Agriculture, Activity Monitoring. Need and features of PLM-IoT, PLM-IoT architecture, tools.

Practical/ Lab work:

At least six Practical assignments on above syllabus.

Reference Books:

- **F. daCosta**, Rethinking the Internet of Things: A Scalable Approach to Connecting Everything, Springer Science+Business Media, New York: Apress Media, LLC, 2013.
- **M. Miller**, Internet of Things The: How Smart TVs Sm: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World, Pearson Education India, 2015.
- **S. Cheruvu**, Demystifying Internet of Things Security: Successful IoT Device/Edge and Platform Security Deployment, Springer Science+Business Media, New York: Apress Media, LLC, 2020.
- **A. Bahga and V. Madiseti**, Internet of Things: A Hands-on Approach, Universities Press, 2015.
- **R. Pethuru and A. C. Raman**, The Internet of Things: Enabling Technologies, Platforms, and Use Cases, CRC Press - Auerbach Publications , 2017.

PRJ-PL521 MINI PROJECT & SEMINAR-II

Objectives of the course:

1. To motivate for using recent technical research papers and articles.
2. To promote for improving learning, presenting and documentation skills.

Course Outcomes:

- a. Apply effective strategies in literature searches using available resources.
- b. Build programs, software tool, interface for automating PLM system issues.
- c. Document properly according to a specific style.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course contents:

The Seminar and Mini Project shall consist of few particulars amongst **literature review** based on a sizable number of publications. **Design / Development / Synthesis** related to a particular area. Implementation of existing theory for applications, pilot experiments etc. Each student is required to prepare a report and deliver a talk based on the work carried out as mini project under the guidance of a faculty member(s). The work carried out should be preferable related to his/her dissertation topic.

OEC-801 BUSINESS ANALYTICS (BA)

Objectives:

1. To understand the role of business analytics within an organization.
2. To analyze data using statistical and data mining techniques and understand relationships between the underlying business processes of an organization.
3. To gain an understanding of how managers use business analytics to formulate and solve business problems and to support managerial decision making.
4. To become familiar with processes needed to develop, report, and analyze business data.
5. To use decision-making tools/Operations research techniques.
6. To manage business process using analytical and management tools.
7. To analyze and solve problems from different industries such as manufacturing, service, retail, software, banking and finance, sports, pharmaceutical, aerospace etc.

Course Outcomes:

At the end of course, students will be able to;

- a) Demonstrate knowledge of data analytics.
- b) Demonstrate the ability of think critically in making decisions based on data and deep analytics.
- c) Demonstrate the ability to use technical skills in predicative and prescriptive modeling to support business decision-making.
- d) Demonstrate the ability to translate data into clear, actionable insights.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Business analytics: Overview of business analytics, scope of business analytics, business analytics process, relationship of business analytics process and organization, competitive advantages of business analytics. statistical tools: statistical notation, descriptive statistical methods, review of probability distribution and data modeling, sampling and estimation methods overview.

Unit II Trendiness and regression analysis: Modeling relationships and trends in data, simple linear regression. important resources, business analytics personnel, data and models for business analytics, problem solving, visualizing and exploring data, business analytics technology.

Unit III Organization structures of business analytics; team management, management issues, designing information policy, outsourcing, ensuring data quality, measuring contribution of business analytics, managing changes. descriptive analytics, predictive analytics, predicative modeling, predictive analytics analysis, data mining, data mining methodologies, prescriptive analytics and its step in the business analytics process, prescriptive modeling, nonlinear optimization.

Unit IV Forecasting techniques: Qualitative and judgmental forecasting, statistical forecasting models, forecasting models for stationary time series, forecasting models for time series with a linear trend, forecasting time series with seasonality, regression forecasting with casual variables, selecting appropriate forecasting models. monte carlo simulation and risk analysis: monte carle simulation using analytic solver platform, new-product development model, newsvendor model, overbooking model, cash budget model.

Unit V Decision analysis: Formulating decision problems, decision strategies with the without outcome probabilities, decision trees, the value of information, utility and decision making.

Unit VI Recent trends in: Embedded and collaborative business intelligence, visual data recovery, data storytelling and data journalism.

Text Books:

- **Business analytics Principles, Concepts, and Applications** by Marc J. Schniederjans, Dara G. Schniederjans, Christopher M. Starkey, Pearson FT Press.
- **Business Analytics** by James Evans, persons Education.

AUD-901 PROJECT MANAGEMENT (PM)

Objectives:

1. To inculcate the knowledge that is required to implement various projects.
2. Develops the vision for identification and formulation of the projects.
3. Demonstrate tools and techniques which are essential for smoother execution of projects.
4. Imparts the knowledge that can be applied to optimize time and resources in project implementation.

Course Outcomes:

At the end of course, students will be able to;

- a) Evaluate feasibility of a project by applying relevant criteria.
- b) Compute deadlines and milestones of a project.
- c) Determine critical path and floats for a project by using CPM.
- d) Compute uncertainties in lifecycle of a project by using PERT.
- e) Perform effective resource allocation in lifecycle of a project.
- f) Determine project and control deadlines while creating Gantt and CPM/PERT chart in Project.

Evaluation scheme:

SN	Component	Weightage (%)
1	Internal semester evaluation I	10
2	Mid Term Evaluation	30
3	Internal semester evaluation II	10
4	End Term Evaluation	50

Course contents:

Unit I Introduction to PM: Projects in Contemporary Organization, Project Life Cycle

Unit II Project Initiation: Strategic Management, Project Selection & Evaluation, Selection Criteria & Models, Risk Management, Portfolio Process, Project Proposals, Project manager: Demands on Project manager, Selecting the Project Manager, Multicultural Communication, Project Organization: Organizational Concepts in PM, Selecting an Organizational Form, Project Planning: Systems integration, WBS & Responsibility Charts, Interface Coordination, Conflict and Negotiation in PM: Nature of Negotiation, Conflict and Project Life Cycle.

Unit III Project Implementation: Budgeting and Cost Estimation: Estimating Project Budgets, Improving Cost Estimation Process, Scheduling: Background, Network Techniques: PERT & CPM, Risk Analysis & Crystal Ball Simulation, Resource Allocation: CPM & Crashing a Project, Resource Allocation, Resource Loading & Levelling, Constrained Resource Scheduling, Multi-project Scheduling & Resource Allocation, Goldratt's Critical Chain, Monitoring & Information System, Planning-Monitoring-Controlling.

Information Needs & Reporting Process, Earned Value Analysis, Computerized PMIS, Project Control: Need for Project Control, Three Types of Control Processes, Design of Control Systems, Control of Creative Activities, Control of Change & Scope, Creep.

Unit IV Project Termination: Project Auditing: System Goals & Project Audit, Audit Report, Project Audit Life Cycle, Project Termination, Varieties of Project Termination, Termination Process, Final Report, A Project History.

Text Books:

- **P. Gopalakrishnan and V. E. Rama Moorthy**, Project Management, Macmillan India Ltd., New Delhi, 1993.
- **Prasanna Chandra**, Projects: Preparation, Appraisal, Budgeting and Implementation, Tata McGraw Hill Publishing Company Ltd., New Delhi, 1980.
- **B. B. Goel**, Project Management: Principles and Techniques, Deep & Deep Publications, New Delhi, 1986.
- **UNIDO Series** on Project Management

DIS-PL601 DISSERTATION - I**Objectives of the course:**

1. To motivate for doing research development activities.
2. To find solutions to realistic industrial problems.
3. To comprehend the skill for report writing, presenting and documentation.

Course Outcomes:

- a. Recognize the importance of planning and preparation required to undertake a research project.
- b. Develop a thorough understanding of the chosen subject area.
- c. Demonstrate the ability to collate and critically assess/interpret data.
- d. Develop an ability to effectively communicate knowledge in a scientific manner.

Evaluation scheme:

Sr. No.	Component	Weightage (%)	Remark
1	In Semester Assessment	30	Continuous Assessment
		20	Mid Semester Assessment
2	End Semester Assessment	50	Assessment based on presentation and demonstration before examiners.
Note: A separate schedule will be displayed for the different activities under Dissertation - I			

Course contents:

The dissertation – I include the following component:

Part Implementation, progress work of the main project.

HSSM-601 INDIAN KNOWLEDGE SYSTEM (IKS)

Objectives: The main objectives of this course are as follows:

1. Creating awareness amongst the youths about the true history and rich culture of the country.
2. Understanding the scientific value of the traditional knowledge of Bharata;
3. Promoting the youths to do research in the various fields of Bharatiya knowledge system.
4. Converting the Bharatiya wisdom into the applied aspect of the modern scientific paradigm.
5. Adding career, professional and business opportunities to the youths.
6. It is also believed that after completion of this course the students will get a holistic insight into the understanding the working of nature and life.

Outcomes:

After completion of course, students will be able to

- CO1. Understand and appreciate the importance of ancient knowledge to a society.
- CO2. Familiarize with the key components of the IKS.
- CO3. Develop some appreciation of IKS historicity.
- CO4. Have an indelible imprint of IKS on mind and soul.
- CO5. Feel the contributions made by ancient Indians in the field of Science, Philosophy and related applications and concepts.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course contents:

Unit 1: Bharatiya Civilization and Development of Knowledge System (4 hours) Genesis of the land, Antiquity of civilization, On the Trail of the Lost River, Discovery of the Saraswati River, the Saraswati Sindhu Civilization, Traditional Knowledge System, The Vedas, Main Schools of Philosophy (6+3), Ancient Education System, the Takshashila University, the Nalanda University, Alumni, Knowledge Export from Bharata.

Unit 2: Arts, Literature and Scholars (4 hours)

Art, Music, and Dance, Nataraja A Masterpiece of Bharatiya Art, Literature, Life and works of Agastya, Lopamudra, Ghosa, Valmiki, Patafijali, Vedavyasa, Yajnavalkya, Gargi, Maitreyi, Bodhayana, Caraka, Susruta, Jivaka, Nagarjuna, Kanada, Patafijali, Kautilya, Panini, Thiruvalluvar, Aryabhata, Varahamihira, Adi Shakaracarya, Bhaskara charya, Madhavacharya.

Unit 3: Science, Astronomy, and Mathematics (4 hours)

Concept of Matter, Life and Universe, Gravity, Sage Agastya's Model of Battery, Velocity of Light, Vimana : Aeronautics, Vedic Cosmology and Modern Concepts, Bharatiya Kalaganana, Kerala School for Mathematics and Astronomy, History and Culture of Astronomy, Sun, Earth, Moon, and Eclipses, Earth is Spherical and Rotation of Earth, Archaeoastronomy; Concepts of Zero and Pi, Number System, Pythagoras Theorem, and Vedic Mathematics.

Unit 4: Engineering, Technology and Architecture (4 hours)

Pre-Harappan and Sindhu Valley Civilization, Laboratory and Apparatus, Juices, Dyes, Paints and Cements, Glass and Pottery, Metallurgy, Engineering Science and Technology in the Vedic Age and Post-Vedic Records, Iron Pillar of Delhi, Rakhigarhi, Mehrgarh, Sindhu Valley Civilization, Marine Technology, and Bet-Dwarka.

Unit 5: Life, Environment and Health (4 hours)

Ethnic Studies, Life Science in Plants, Anatomy, Physiology, Agriculture, Ecology and Environment, Ayurveda, Integrated Approach to Health care, Medicine, Microbiology, Medicine, Surgery, and Yoga, etc.

References:

1. Textbook on The Knowledge System of Bharata by Bhag Chand Chauhan,
2. History of Science in India Volume-1, Part-I, Part-II, Volume VIII, by Sibaji Raha, et al., National Academy of Sciences, India and The Ramkrishan Mission Institute of Culture, Kolkata (2014).
3. Pride of India-A Glimpse of India's Scientific Heritage edited by Pradeep Kohle et al. Samskrit Bharati (2006).
4. Vedic Physics by Keshav Dev Verma, Motilal Banarsidass Publishers (2012).
5. India's Glorious Scientific Tradition by Suresh Soni, Ocean Books Pvt.Ltd. (2010).

DIS-PL602 DISSERTATION - II

Objectives of the course:

1. To motivate for doing research development activities.
2. To find solutions to realistic industrial problems.
3. To comprehend the skill for report writing, presenting and documentation.

Course Outcomes:

- a. Recognize the importance of planning and preparation required to undertake a research project.
- b. Extend the research work to a standardized level that can give solutions to real life issues.
- c. Demonstrate the ability to collate and critically assess/interpret data.
- d. Develop an ability to effectively communicate knowledge in a scientific manner.

Evaluation scheme:

Sr. No.	Component	Weightage (%)	Remark
1	In Semester Assessment	30	Continuous Assessment
		20	Mid Semester Assessment
2	End Semester Assessment	50	Viva-Voce

Note: A separate schedule will be displayed for the different activities under Dissertation - II

Contents:

The dissertation work shall consist of an extensive work, study or analysis of field / industrial problems with appropriate solutions or remedies. The bonafide work carried out for Dissertation – II should be potentially rich in terms of academics.

Dissertation Report

The project report shall be hard bound. It is a report on the work done by the student. It should have literature review, problem definition and formulation, adopted methodology, experimentation plan if any, results, conclusions, discussion and its relevance to the further work.

Examination

The viva-voce examination of the Dissertation – II shall consist of a presentation by the candidate and demonstration of the work carried out.

CCC-602 YOGA AND SPORTS

Objectives: The main objectives of this course are as follows:

1. To make the students understand the importance of sound health and fitness principles as they relate to better health.
2. To expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness.
3. To create a safe, progressive, methodical and efficient activity-based plan to enhance improvement and minimize risk of injury.
4. To develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Outcomes: After completion of course, students will be able to

CO1. To practice Physical activities and Hatha Yoga focusing on yoga for strength, flexibility, and relaxation.

CO2. To learn techniques for increasing concentration and decreasing anxiety which leads to stronger academic performance.

CO3. To learn breathing exercises and healthy fitness activities

CO4. To perform yoga movements in various combination and forms. COS. To assess current personal fitness levels.

CO5. To demonstrate an understanding of sound nutritional practices as related to health and physical performance.

Evaluation scheme:

Sr. No.	Component	Weightage (%)
1	Mid Term Evaluation	50
2	End Term Evaluation	50

Course contents:

Unit 1: Introduction to Physical Education (1 hour)

Meaning and definition of Physical Education

- Aims and objectives of Physical Education
- Changing trends in Physical Education

Unit 2: Olympic Movement (2 hours)

Ancient and Modern Olympics (Summer and Winter) Olympic Symbols, Ideals, Objectives and Values

Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc.)

Unit 3: Physical Fitness, Wellness & Lifestyle (2 hours) Meaning and Importance of Physical Fitness & Wellness Components of Physical fitness

Components of Health-related fitness Components of wellness

- Preventing Health Threats through Lifestyle Change
- Concept of Positive Lifestyle

Unit 4 : Fundamentals of Anatomy & Physiology in Physical Education, Sports and Yoga (2 hours)

Define Anatomy, Physiology & Its Importance

- Effect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.)

Unit 5: Kinesiology, Biomechanics & Sports (2 hours)

Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports

- Newton's Law of Motion & its application in sports.
- Friction and its effects in Sports.

Unit 6: Postures (1 hour)

Meaning and Concept of Postures. Causes of Bad Posture.

Advantages & disadvantages of weight training. Concept & advantages of Correct Posture.
Common Postural Deformities - Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow
Legs and Scoliosis.

Corrective Measures for Postural Deformities

Unit7: Yoga (1 hour)

Meaning and Importance of Yoga Elements of Yoga

Introduction -Asanas, Pranayama, Meditation & Yogic Kriyas

Yoga for concentration and related Asanas (Sukhasana; Tadasana; Padmasana &
Shashankasana)

Relaxation Techniques for improving concentration- Yog-nidra.

Unit 8: Yoga and Lifestyle (2 hours)

Asanas as preventive measures.

Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana.

Obesity: Procedure, Benefits and contraindications for Vajrasana, Hastasana, Trikonasana,
Ardh Matsyendrasana.

Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana.

Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan
Muktasana, Ardh Matsyendrasana.

Asthma: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana,
Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.

Unit 9: Training and Planning in Sports (1 hour)

Meaning of Training.

Warming up and limbering down Skill, Technique & Style.

Meaning and Objectives of Planning.

Tournament - Knock-Out, League/ Round Robin and Combination.

Unit 10: Psychology & Sports (1 hour)

Definition & Importance of Psychology in Physical Edu. & Sports Define & Differentiate Between
Growth & Development Adolescent Problems & Their Management.

Emotion: Concept, Type & Controlling of emotions Meaning, Concept & Types of Aggressions in
Sports. Psychological benefits of exercise.

Anxiety & Fear and its effects on Sports Performance. Motivation, its type & techniques.

Understanding Stress & Coping Strategies.

Unit 11: Doping (1 hour)

Meaning and Concept of Doping Prohibited Substances & Methods Side Effects of Prohibited
Substances.

Unit 12: Sports Medicine (01 hour)

First Aid - Definition, Aims & Objectives.

- Sports injuries: Classification, Causes & Prevention.
- Management of Injuries: Soft Tissue Injuries and Bone & Joint Injuries.

**Unit 13: Sports/ Games Following subtopics related to any one Game/Sport of choice of student out of:
Athletics, Badminton, Basketball, Chess, Cricket, Kabaddi, Lawn Tennis, Swimming, Table Tennis,
Volleyball, Yoga etc.**

History of the Game/Sport.

Latest General Rules of the Game/Sport.

Specifications of Play Fields and Related Sports Equipment. Important Tournaments and Venues.

Sports Personalities. Proper Sports Gear and its Importance.

References:

1. Modern Trends and Physical Education by Prof. Ajmer Singh.
2. Light on Yoga By B.K.S. Iyengar.
3. Health and Physical Education - NCERT (11th and 12th Classes)

List of Experiments: Students have to practice different Asanas mentioned in the syllabus, study different sports.